Project Title: Cloud-Based Student Attendance Management System

Student Name(s): _____ **Course:** Cloud Computing — Module 4

Objective

To develop a simple cloud-ready application that records student attendance and generates date-wise attendance summaries, reducing manual effort and improving record accuracy.

Problem Statement

Manual attendance tracking can be slow, difficult to maintain, and prone to recording errors. A centralized web application makes attendance entry and reporting more accessible.

Methodology / Tools Used

Python and Flask are used to implement the web application and HTTP routes. SQLite stores attendance entries, while HTML/CSS provide the user interface. Gunicorn and deployment configuration support hosting on a cloud platform such as Render; Docker is included as an optional containerization method.

Implementation

Users enter a student's name, ID, attendance date, and status (Present/Absent). The system validates required fields and prevents duplicate records for the same student on a given date. A report page filters entries by date and displays present, absent, and total counts. A health endpoint is provided for basic service checking.

Results

The prototype successfully records attendance and produces a date-specific summary through a browser. It demonstrates a cloud programming workflow: web interface, application logic, data persistence, deployment configuration, and access through a hosted URL.

Conclusion

The project meets its objective of demonstrating a practical cloud-ready application. It can be extended with authentication, class-wise attendance, CSV export, and a managed cloud database. SQLite is intended for demonstration; persistent managed storage is recommended for production deployment.

References

1. Flask Documentation — https://flask.palletsprojects.com/ 2. Python Documentation — https://docs.python.org/3/ 3. SQLite Documentation — https://www.sqlite.org/docs.html 4. Render Documentation — https://render.com/docs